

Print at 100% / Actual size, SINGLE-sided. Crease along the grey line, fold so both printed faces are outward, glue, then cut. This bar must measure exactly 80 mm.

[illegible]

FIRST SOLVE

New here? Do the walk-through once on [cubepath3.vserved.app/learn](#). This card is the memory jog.

NOTATION
R U **F** L **D** B = turn that face clockwise, looking at from outside the cube.
r = counter-clockwise. **2** = half turn. Lowercase **r** = that face plus the layer behind it, turning together.
y = spin the whole cube left, white staying down. A whole-cube turn solves nothing.

WORDS
 algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out.

Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top. Turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
RUR'U'



white sticker FRONT
L'L'LU



white sticker UP run the first one once to knock it out, then look again
RUR'U'

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center inside it. The arrow shows which slot it goes to.



goes RIGHT
U **RU** **y** **L'L'LU**



goes LEFT
U **R'U'** **L'L'LU** **y'** **RUR'U'**

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

[illegible]

TWO-LOOK

OLL CROSS - yellow edges



Line bar left-front right
fRUR'U'f



Hook hook pointing front-right
fRUR'U'f'

OLL CORNERS - yellow face



Sune $\pm$ 1 yellow; rest clockwise
RUR'U' RU2R'



Anti-Sune $\pm$ 1 yellow; rest counter-clockwise
RU2R' U'RUR'



Pi $\pm$ 0 yellow; headlights left
fRUR'U'f' fRUR'U'f'

OLL CORNERS

- continued



Headlights $\pm$ 2 yellow; headlights at the back
RD2R'U2' RD'R'U2R'



2x Headlights $\pm$ 0 yellow; headlights both sides
RUR'U' RUR'R' URU2R'



Chameleon $\pm$ 2 yellow; net to each other
rRUR'U' FRF'



Bowtie $\pm$ 2 yellow diagonal
F'RUR'U' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn, also, finishes the case. Machine-checked, there is the worst. Top not all yellow and nothing matches? Run Sun and read again.

No yellow edge at all? Run the first one, then read again.

RUR'U' sexy move **RUR'U'** Sun **R'FRF'** algachemist *gap* = change grip

[illegible][illegible]

ONE-LOOK PLL ADJACENT CORNER SWAP — exactly one face shows headlights  - T L headlights; pairs B-left F-left RUR'U' R'F R2UR'UR' RUR'F' - J a solid; pairs B-right F-left B-front x R2FRF' UR2 r'Ur U2 - J b L solid; pairs B-left F-back RUR'F' RUR'U' R'F R2UR'U' - A a L headlights; pairs F-right B-front x LZD2 L'UL D2 L'UL' - A b L headlights; pairs B-right B-back x' LZD2 LUL' D2 LU'L - F I solid RUR'U' RUR'U' R'F R2UR'U' RUR'UR One face shows headlights: two corners of that face match. Hold as drawn, run, then turn the top to finish. RUR'U' UR UR'U' R' FRF' s'dolorchomoc — can = change, s'tro	ADJACENT CORNER SWAP — continued  - R a L headlights; pairs R-left Ra'R'U' RUR DR'UR'D' R'U2R' - R b L headlights; pairs B-left Rb'L RURUR' F'R U2R'U2R - G a L headlights; pairs F-right R2UR'UR'U' RU'R2 U'D R'UR'D' - G b L headlights; pairs R-back RUR'UR' UD' R2UR'UR' RU'R2D - G c L headlights; pairs B-right R2UR'RU'R UR'R2 UD' RU'R'D - G d L headlights; pairs R-front RUR' UD' R2UR'UR' UR'UR2D' Learn in the printed order, three a week, about six weeks: Ja & Jb on the same day — one is the mirror of the other.
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BIG CUBES – BIG CUBES AND THE MAP

BIG CUBES – 4x4 and 5x5
 Reduce; build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers deep. 2Rw = 2nd Layer ONLY – never invert it.
last 2 edges

$$\begin{matrix} & \text{R}'\text{w} & \text{U}' & \text{R}' & \text{U} & \text{F} & \text{R}' & \text{F} \\ 2\text{R}'2 & \text{U}'2 & 2\text{R}'2 & \text{U}'2 & 2\text{R}'2 & \text{U}'2 & & \end{matrix}$$
 edge pair, run it twice, slice back – both cubes
 slice pairs swapped - half the time

4x4 PLL parity
 4x4 OLL parity = 1 edge gap flipped - half the time

$$\begin{matrix} \text{Rw} & \text{U}'2 & \text{x} & \text{Rw} & \text{U}'2 & \text{R}'2 & \text{U}'2 & \text{L}'w & \text{U}'2 & \text{R}'2 & \text{Rw} & \text{U}'2 & \text{R}'w' & \text{U}'w' \\ 5x5 \text{ edge parity} & 1 \text{ edge gap flipped - half the time - one move differs. } 5x5 \text{ has no PLL parity} \\ \text{Rw} & \text{U}'2 & \text{x} & \text{Rw} & \text{U}'2 & \text{R}'2 & \text{U}'2 & \text{L}'w &quad \text{5x5} & \text{U}'2 & \text{R}'2 & \text{Rw} & \text{U}'2 & \text{R}'w' & \text{U}'w' \end{matrix}$$
 last layer, during the last layer, both parity algorithms more corners

NOTATION
 Each letter turns that face clockwise,
 seen from outside the cube – so from your seat L and B look counter-clockwise → counter-clockwise = 2 half turn.
 M = middle slice, turning the way L turns. x y z = whole cube, turning the way R/U/B turn. Lowercase f = that face plus the layer behind it, turning together.



BEGINNER SHORTCUT - use it until you know Sure
 Twist a yellow corner four turns at a time, turning only the top face between corners.
yellow faces RIGHT $\begin{matrix} \text{D}'\text{R}'\text{D} & \times 2 \\ \text{R}'\text{D}'\text{R} & \times 2 \end{matrix}$
yellow faces FRONT